

1.1)

$$35) |a - b| \geq |a| - |b|$$

DICA:

$$|a| = |(a - b) + b|$$

←

(1.5)

$$|A + B| \leq |A| + |B|$$

$$\underline{= |a|} = | \underbrace{(a - b)}_A + b | = |A - b|$$

$$\stackrel{1.5}{\leq} |A| + |b|$$

$$\leq \underline{\underline{|a - b| + |b|}}$$

$$\text{Ou seja, } |a| \stackrel{-|b|}{\leq} |a - b| + \cancel{|b|} \stackrel{-|b|}{-|b|}$$

$$|a| - |b| \leq |a - b|$$

$$36) |a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|$$

$$|a_1 + \underbrace{a_2 + \dots + a_n}_B| \stackrel{1.5}{\leq} |a_1| + |a_2 + \dots + a_n|$$

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + \underbrace{|a_2 + \dots + a_n|}_{\text{⊗}}$$

$$|a_2 + \underbrace{a_3 + \dots + a_n}_B| \leq |a_2| + |a_3 + \dots + a_n|$$

$$\Rightarrow |a_1| + |a_2 + a_3 + \dots + a_n| \leq |a_1| + |a_2| + |a_3 + \dots + a_n|$$

PORTANTO:

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \underbrace{|a_3 + \dots + a_n|}_A$$

RACIOCÍNIO RECURSIVO.

Se CONTINUARMOS...

$$\leq |a_1| + |a_2| + |a_3| + \dots + |a_n|$$

37] Hipótese $0 < a < b$, então TESE $\frac{1}{a} > \frac{1}{b}$

$0 < \overset{\times \frac{1}{a}}{a} < \overset{\times \frac{1}{a}}{b} \Rightarrow a \cdot \frac{1}{a} < b \cdot \frac{1}{a}$

$\Rightarrow 1 < \frac{b}{a} \Rightarrow 1 \cdot \frac{1}{b} < \frac{b}{a} \cdot \frac{1}{b}$

$\Rightarrow \frac{1}{b} < \frac{1}{a}$

ou

$\frac{a}{ba} < \frac{b}{ba}$

$\frac{1}{b} < \frac{1}{a}$

38] Hipótese $0 < a < b \Rightarrow$ TESE $a^2 < b^2$

$a < b \Rightarrow a \cdot (a+b) < b \cdot (a+b)$

$a^2 + \cancel{ab} < \cancel{ba} + b^2$

$a^2 < b^2$

$a < b \Rightarrow a^2 < \underline{b \cdot a}$

$a < b \Rightarrow \underline{ab} < b^2$

$a^2 < ba = ab < b^2$

$a^2 < b^2$

39] $a < b$

$c < d$

$\Rightarrow a + c < b + d$

Hip.

$$\begin{array}{l} a < b \Rightarrow \underline{a+c} < b+c \\ c < d \Rightarrow b+c < \underline{b+d} \end{array} \Rightarrow a+c < b+d$$

40] $a < b$
 $c < d$

 Hip.

$$ac < bd$$

(?)

$$\left. \begin{array}{l} a = -10 \text{ e } b = -5 \\ c = 1 \text{ e } d = 2 \end{array} \right\} \begin{array}{l} -10 \cdot 1 < -5 \cdot 2 \\ -10 < -10 \\ \text{(ABSURDO)} \end{array}$$

Obs: Porém se ...

$$\left. \begin{array}{l} b > a > 0 \\ d > c > 0 \end{array} \right\} \textcircled{M}$$

⇓

$$\underline{cb} > ac$$

e

$$\underline{bd} > \underline{bc}$$

$$bd > bc = cb > ac$$

$bd > ac$